Sleep and recovery

The most underrated lever for performance, fat loss and health. Sleep badly and you sabotage even a perfect diet and a solid training plan. Here is what to do about it. Practical, and based on the research.

Why sleep matters this much

Sleep is not "time when nothing happens". It is the most active recovery window of the day: muscles get repaired, memory gets stored, hunger and stress hormones get tuned. Cut it short and several things drop at once:

- ✓ **More hunger, more cravings for sweet food.** Just two nights of 4 hours of sleep raise hunger by about a quarter: ghrelin (hunger) goes up, leptin (fullness) goes down, and you want calorie-dense food more [1,2].
- ✓ **In a deficit you lose more muscle than fat.** In a study with the same calorie deficit, short sleepers (5.5 h) lost about 55% less fat and about 60% more muscle than people who slept 8.5 h [3]. Same diet, different sleep. Different result.
- ✓ **Worse performance and recovery.** Less sleep = less strength, endurance and accuracy, and a higher injury risk. Extending sleep, on the other hand, improves performance in athletes [4,5].
- ✓ **Weaker willpower and worse decisions.** A tired brain resists temptation less well and makes impulsive food choices [2].

> **Treat sleep as part of the training plan**, not as a reward for when you have time. It is the cheapest "supplement" you have.

How much, and how well

For an adult the recommendation is **7-9 hours** [6]. One long night matters less than **consistency**: go to bed and get up at a similar time, including the weekend. Irregular sleep (so-called social jet lag) is linked with worse metabolic health on its own. Watch three things: **how much** you sleep, **how regular** it is, and **how you feel** after waking and around 15:00 (an alertness dip there is normal, but it should not flatten you).

7 steps to better sleep

1 A fixed wake-up time

Get up at the same time every day, even after a bad night. A fixed morning anchor sets the rhythm better than the time you fall asleep.

2 Light right after waking

5-15 min of daylight (outside or by a window) after you wake up. It strengthens morning alertness and evening sleepiness through the circadian rhythm [7].

3 Dark and cool

Keep the bedroom dark and on the cool side (about 18-19 °C, roughly 64-66 °F). A drop in body temperature helps sleep start. Blackout or a sleep mask does a lot.

4 Caffeine with a cutoff

Last coffee about 8-10 h before bed. Caffeine's half-life is about 5 h [8]. An afternoon espresso can still bother you at midnight.

5 Screens and stimulation down

For 30-60 min before bed, dim the lights and put the phone away. Mental stimulation and the time you lose do more damage than "blue light" itself.

6 Alcohol is not a sleep aid

You fall asleep faster, but the second half of the night is light and broken up (less REM) [9]. You wake up wrecked.

7 A shutdown routine

A few minutes of the same ritual (shower, reading, breathing at 4-6 breaths/min). A signal to the body: we are going to sleep.

Supplements: what is worth it

Fix the routine first, then the pills. If you still want support:

- ✓ **Melatonin (0.5-3 mg):** helps most when your rhythm is shifted (jet lag, night shifts, falling asleep too late). It shortens the time to fall asleep a little. It is not a sleeping pill [10]. Take it about 1-2 h before bed.
- ✓ **Magnesium:** reasonable mainly when your food is low in it. Don't expect miracles.
- ✓ **Glycine, L-theanine:** a milder calming effect for some people. Safe, but optional.
- ✓ **Watch "sleep stacks" with high doses**, and don't take sleep medication as a routine without a doctor.

When it still doesn't work

- ✓ Can't fall asleep for 20+ min? Get up, do something boring in dim light, and go back when you feel sleepy. Don't toss and turn in bed.

- ✓ Awake at 3 a.m. with a busy mind? Write the thoughts down in the evening, and cut alcohol and a heavy meal late at night.
- ✓ Stress won't let go? Try 5 minutes of slow breathing (longer exhale) or a short "brain dump" of tomorrow's tasks.
- ✓ **Snoring, gasping for air, extreme daytime tiredness?** That belongs with a doctor. It can be sleep apnea, and sleep hygiene alone will not fix it.

> **Practical tip:** Pick **one** of the 7 steps, the one that takes the least effort, and keep it for 7 days. When it runs by itself, add a second. Small steps last. A huge resolution fades.

© Martin Barna: client guide. Share only with the author's permission. **martinbarna.cz/en**

Sources: [1] Spiegel K et al. Ann Intern Med. 2004;141(11):846-50. [2] Greer SM et al. Nat Commun. 2013 (sleep and appetite). [3] Nedeltcheva AV et al. Ann Intern Med. 2010;153(7):435-41. [4] Fullagar HHK et al. Sports Med. 2015;45(2):161-86. [5] Mah CD et al. Sleep. 2011;34(7):943-50. [6] Hirshkowitz M et al. Sleep Health. 2015;1(1):40-43 (National Sleep Foundation). [7] Blume C et al. Somnologie. 2019 (light and circadian rhythm). [8] Institute of Medicine, caffeine pharmacology (half-life about 5 h). [9] Ebrahim IO et al. Alcohol Clin Exp Res. 2013 (alcohol and sleep architecture). [10] Ferracioli-Oda E et al. PLoS One. 2013 (melatonin meta-analysis). Overview: Walker M. Why We Sleep. 2017.